A Simulation Framework for Restaurant Reservation Recommendation under Capacity Constraints

Abstract

This study addresses the problem of retaining customers whose original restaurant reservation requests cannot be accommodated because of capacity constraints. It proposes a simulation framework for evaluating adaptive strategies for recommending alternative reservation offers. The framework integrates restaurant capacity and occupancy, heterogeneous customer preferences, feasible alternative offer generation, and probabilistic customer decision-making. Several multi-armed bandit algorithms are compared with random offer selection under different restaurant operating conditions and customer behavior scenarios. The results show that adaptive strategies generally outperform random selection, particularly when only a small number of alternatives is displayed. However, the differences among the evaluated multi-armed bandit algorithms were generally small, suggesting that the design and composition of the alternative offers have a greater influence on reservation recovery than the choice of a particular adaptive algorithm. In the baseline scenario, the decay ε-greedy strategy achieved an approximately 19.5% relative improvement in the reservation recovery rate over random selection when only one alternative was displayed. Increasing the number of displayed alternatives from one to seven raised the recovery rate from 15.49% to 52.57%, with most of the improvement occurring within the first four to five offers. Customer decision uncertainty and initial attitude substantially influenced the relative benefit of adaptive recommendation, while offers involving financial incentives achieved the highest acceptance rates. The proposed framework enables the systematic evaluation of recommendation strategies and alternative offer designs under varying conditions before their potential deployment in real-world restaurant reservation systems.

Keywords: restaurant reservation systems; simulation framework; multi-armed bandits; recommender systems; capacity-constrained services; adaptive recommendation

1 Introduction

Restaurant reservation systems play an important role in restaurant operations. They facilitate capacity planning and operational coordination for restaurant operators while providing customers with greater certainty regarding the availability of the requested service. During periods of high demand, however, restaurants may be unable to accommodate an original reservation request and may therefore have to reject it. Such a situation does not necessarily have to result in the loss of a customer if the reservation system can provide a suitable alternative, such as a different reservation time, a financial incentive, or a non-financial benefit. The key challenge is therefore to select an alternative offer that the customer is willing to accept despite the negative experience associated with the rejection of the original request.

The strategy used to select alternative offers represents an important factor affecting their probability of acceptance. However, there is no clearly defined procedure or universally optimal approach for selecting and ranking individual alternatives. Moreover, recommendation performance may depend not only on the selection strategy but also on the number, type, feasibility, and diversity of the available alternatives, as well as on restaurant operating conditions and

customer decision-making behavior. There is therefore a need to systematically compare different selection strategies while also assessing the relative importance of the environment in which these strategies operate.

The selection of an alternative offer can be viewed as a sequential decision-making problem in which the system initially does not know the acceptance probabilities of the individual options. Through repeated interactions with customers, however, it can use the feedback obtained to gradually adapt its decisions. Multi-armed bandit algorithms are suitable for this type of problem because they balance the exploitation of offers that have performed well in the past with the continued exploration of less frequently tested alternatives. Consequently, they can respond to differences between individual offer types without requiring precise prior knowledge of customer preferences.

This study proposes a simulation framework for evaluating adaptive strategies for recommending alternatives to rejected reservation requests. The framework enables several multi-armed bandit algorithms to be compared with random offer selection under different restaurant operating conditions and customer behavior scenarios. The analysis focuses on the effects of the recommendation strategy, the number and type of displayed alternatives, restaurant occupancy, customer decision uncertainty, and customers' initial attitudes toward the proposed alternatives.

The objective of the study is to determine whether adaptive selection strategies outperform random selection, examine the extent to which individual adaptive strategies differ from one another, identify which characteristics and categories of alternative offers contribute most to customer acceptance, and assess how recommendation performance changes under different environmental parameters and restaurant operating conditions. In addition to identifying the best-performing strategies, the study therefore examines whether the choice of a particular learning algorithm has a greater influence on reservation recovery than the design and composition of the available alternative offers.

Based on the defined problem, the study addresses the following research questions:

RQ1: Number of Displayed Offers. How does the number of displayed alternatives affect the probability that a customer accepts an offer?

RQ2: Alternative Offer Categories. Which categories of alternative offers achieve the highest acceptance rates?

RQ3: Robustness under Different Operating Conditions. How do restaurant operating conditions and customer decision-making behavior affect the effectiveness of adaptive recommendation?

RQ4: Comparison of Recommendation Strategies. How do the evaluated multi-armed bandit strategies perform relative to random selection and to one another, and how does their relative performance depend on the number of displayed offers?

The main contribution of this study is the implementation of a simulation framework that enables the systematic evaluation of different recommendation strategies and alternative offer designs. The framework integrates a model of restaurant capacity and occupancy, the generation of customers with heterogeneous preferences, the creation of feasible alternative offers, and a probabilistic model of customer decision-making. It therefore makes it possible to evaluate not only the overall reservation recovery rate but also the acceptance rates of individual offer categories, the relative improvement achieved by adaptive strategies over random selection, and the differences among individual adaptive strategies.

By modifying the model parameters, different restaurant operating conditions and patterns of customer decision-making can be simulated, making it possible to identify the situations in which adaptive recommendation provides the greatest benefit. Importantly, the framework also

enables the relative influence of the recommendation algorithm and the composition of the candidate offer set to be assessed. This distinction is practically relevant because improvements in recommendation performance may depend more on generating feasible, relevant, and sufficiently diverse alternatives than on selecting a more sophisticated learning algorithm. The findings may therefore support both the design and personalization of alternative offers and the selection of suitable recommendation strategies before their validation and potential deployment in real-world restaurant reservation systems.

2 Related Work

2.1 Restaurant Reservation Systems

Restaurant reservation systems have become an essential tool for managing restaurant capacity and coordinating customer demand by scheduling table occupancy and customer arrivals. Unlike reservation systems in accommodation services, restaurant reservations are characterized by substantial uncertainty regarding the duration of table occupancy. Reservation length depends on several factors, including party size, dining duration, and customer behavior, making capacity planning considerably more challenging than in environments with fixed reservation durations (Thompson, 2015, 2019).

Previous research has therefore focused on developing reservation models that simultaneously balance efficient capacity utilization and customer service quality. Since customers generally expect their reserved table to be available at the scheduled time and perceive waiting negatively, reservation systems must carefully manage operational uncertainty while minimizing customer inconvenience. One important research direction has been the incorporation of customer time flexibility into reservation planning. Rather than assuming that customers require a single reservation time, several studies have shown that offering alternative reservation slots can substantially improve restaurant capacity utilization while maintaining acceptable service quality (Thompson, 2019).

Another major research area concerns reservation disruptions, particularly no-shows, late cancellations, overbooking, and walk-in customers. These factors reduce the accuracy of demand forecasting and may result in inefficient capacity utilization. Consequently, reservation models increasingly incorporate these sources of uncertainty with the objective of maximizing expected restaurant revenue while ensuring reliable customer service (Tse and Poon, 2016).

More recently, research has increasingly emphasized behavioral aspects of restaurant reservation systems. Studies have demonstrated that cancellation policies, the complexity of the cancellation process, and customer communication strategies significantly influence reservation behavior and no-show rates (Kim, Tang, Kim, and Kim, 2025). These findings highlight that successful reservation management requires not only effective allocation of limited restaurant capacity but also a thorough understanding of customer decision-making processes.

Overall, existing research demonstrates that restaurant reservation systems represent a joint operational and behavioral optimization problem. Effective reservation management therefore requires simultaneously considering restaurant capacity constraints and customer behavior when designing reservation policies.

Although previous research has extensively investigated reservation scheduling, customer time flexibility, and reservation disruptions such as no-shows and cancellations, considerably less attention has been devoted to the recommendation of alternative reservation offers after an original reservation request cannot be accommodated. In particular, little research has examined how adaptive recommendation methods can be employed to personalize alternative reservation offers under dynamic operating conditions. This research gap motivates the simulation framework proposed in this study.

2.2 Restaurant Revenue Management

Revenue management comprises a set of analytical and operational methods aimed at forecasting customer demand and optimizing the allocation of limited capacity in order to maximize revenue. Its fundamental objective is to provide the right product to the right customer at the right time and at the right price (Kimes, 1989; Tyagi and Bolia, 2021).

Originally developed for the airline and hotel industries, revenue management has gradually expanded to other service sectors characterized by perishable capacity and fluctuating demand, including restaurants (Kimes, 1989; Tyagi and Bolia, 2021). Contemporary restaurant revenue management extends beyond pricing and capacity allocation to encompass reservation management, customer arrivals, customer experience, and digital distribution channels. Recent literature further identifies reservation management, overbooking, no-shows, demand forecasting, and AI-driven decision support as key research directions in the continued development of restaurant revenue management (Kimes, 1989; Tyagi and Bolia, 2021).

An important aspect of restaurant revenue management is the heterogeneity of customer demand. Customers differ not only in their preferences but also in the economic value they generate for the restaurant. Wang, Lin, Kuo, and Weng (2017) argue that commonly used indicators, such as party size, do not necessarily reflect the true value of a reservation. Consequently, effective reservation management should account for differences between customer segments and their impact on both capacity utilization and expected revenue.

Another frequently investigated topic concerns actively influencing customer behavior through various incentives. Previous studies have shown that customers are willing to modify their reservation time in exchange for price discounts or other benefits, enabling restaurants to shift demand away from peak periods and improve capacity utilization (Kimes, 1989; Tyagi and Bolia, 2021). Consequently, restaurant revenue management is concerned not only with responding to existing demand but also with actively shaping customer behavior through appropriately designed incentives.

Recent review studies indicate a growing shift toward data-driven restaurant revenue management based on advanced analytics, predictive modeling, and artificial intelligence. Increasing attention is being devoted to the use of customer data, machine learning techniques, and personalized decision support for improving demand management and revenue optimization (Bujalance-López, González-Serrano, Lechuga Sancho, and Talón-Ballestero, 2025). Although these developments demonstrate the growing role of intelligent decision support in restaurant operations, relatively little attention has been devoted to adaptive recommendation of alternative reservation offers following unsuccessful reservation requests.

2.3 Recommender Systems

Recommender systems are designed to identify and rank relevant alternatives for individual users with the objective of supporting decision-making in situations involving a large number of available options (Adomavicius and Tuzhilin, 2005). By filtering and prioritizing candidate items according to estimated user preferences, recommender systems reduce the cognitive effort required to evaluate all possible alternatives.

An important research direction concerns constraint-based recommender systems, which explicitly incorporate user requirements together with domain-specific constraints into the recommendation process. Rather than identifying the highest-rated item from the entire catalog, these systems generate recommendations that simultaneously satisfy predefined feasibility constraints. Such approaches are particularly suitable for domains involving complex decision-making, including financial products, technical product configuration, travel planning, and educational planning (Felfernig, Friedrich, Jannach, and Zanker, 2015).

Related challenges arise in recommendation problems involving multiple simultaneous constraints. Parameswaran, Venetis, and Garcia-Molina (2011) argue that filtering recommenda-

tions after candidate generation often leads to suboptimal results and that domain constraints should instead be incorporated directly into the recommendation process. This enables the generation of recommendations that satisfy both user preferences and operational requirements.

Besides selecting feasible alternatives, recommender systems must also determine the order in which recommendations are presented. Modern recommender systems typically return a ranked list of alternatives rather than a single recommendation. Consequently, recommendation is often formulated as a learning-to-rank problem, where the objective is to optimize the ordering of items and maximize the relevance of the top-k recommendations presented to the user (Karatzoglou, Baltrunas, and Shi, 2013).

Although recommender systems provide effective mechanisms for selecting and ranking feasible alternatives, they generally assume that user preferences are known or can be estimated from historical interactions. In many real-world decision problems, however, customer preferences must be learned online while recommendations are simultaneously generated. This exploration–exploitation trade-off motivates the use of online learning methods, particularly Multi-Armed Bandit algorithms, which are introduced in the following section.

2.4 Multi-Armed Bandits

Multi-Armed Bandits (MAB) represent a sequential decision-making framework in which an agent repeatedly selects one of several available actions and receives feedback only for the chosen action. The central challenge is to balance *exploration*, i.e., gathering information about uncertain alternatives, with *exploitation*, i.e., selecting alternatives that have previously produced high rewards (Sutton and Barto, 2018).

The exploration–exploitation dilemma naturally arises in recommender systems, where user preferences are initially unknown and must be learned through repeated interactions. Li, Chu, Langford, and Schapire (2010) demonstrated that contextual bandit algorithms can effectively personalize news recommendations by continuously updating estimates of user preferences based on observed feedback. Similar principles can be applied to restaurant reservation systems, where alternative reservation offers correspond to available actions and customer acceptance provides the reward signal.

Several exploration strategies have been proposed to address the Multi-Armed Bandit problem. The ε-greedy strategy selects the currently best-performing action with high probability while occasionally exploring random alternatives. A common extension gradually decreases the exploration rate during learning to shift from exploration to exploitation over time (Bubeck and Cesa-Bianchi, 2012; Sutton and Barto, 2018). Softmax (Boltzmann exploration) assigns selection probabilities according to the estimated rewards of individual actions, enabling smoother exploration than purely random selection (Kaelbling, Littman, and Moore, 1996). Upper Confidence Bound (UCB1) methods explicitly incorporate uncertainty into action selection by combining expected reward estimates with confidence intervals (Auer, Cesa-Bianchi, and Fischer, 2002). Thompson Sampling follows a Bayesian approach, selecting actions according to the probability that they are optimal given the available observations (Thompson, 1933).

Recent studies further suggest that the effectiveness of Multi-Armed Bandit algorithms depends not only on the algorithm itself but also on the characteristics of the decision environment and user behavior. In particular, violations of common modeling assumptions, such as stationary user preferences, may substantially influence algorithm performance in practical recommender systems (Leqi, Zhou, Kılınç-Karzan, Lipton, and Montgomery, 2023).

Although Multi-Armed Bandit methods have been successfully applied in recommender systems and other sequential decision-making problems, their application to adaptive recommendation of alternative restaurant reservation offers has received little attention in the existing literature. This study addresses this gap by evaluating several widely used bandit strategies within a simulation framework designed for restaurant reservation systems.

3 Simulation Framework

The proposed simulation framework models the process of recommending alternative offers after the rejection of a customer's reservation request due to capacity constraints. The framework captures the interaction between restaurants, the customer, alternative offers, and the customer's decision-making process. The simulation is designed to evaluate the performance of various strategies to recommend alternatives under changing operational conditions.

3.1 Problem Definition

The addressed problem considers a situation in which a restaurant is unable to accommodate a customer's original reservation request. Instead of immediately rejecting the reservation, the reservation system attempts to retain the customer by offering one or more alternative options. The goal is to propose an alternative offer that the customer will accept despite the change in the originally requested reservation conditions. Alternative offers are selected based on a defined recommendation strategy and presented to the customer sequentially. Alternatives may include, for example, a time shift, a discount, or a non-financial benefit. The goal of the simulation is to maximize the probability that the customer will accept one of the offered alternatives. The simulation begins once the customer's original reservation has already been declined. The system therefore does not handle the initial restaurant selection, the reservation process, or demand forecasting.

3.2 Framework Architecture

The proposed framework consists of several interconnected components that together model the process of recommending alternative booking options. The overall simulation process is illustrated in Figure 1.

The simulation begins by creating a synthetic customer and assigning them to a restaurant that is known in advance to be unable to accommodate their original reservation request. Based on the restaurant's characteristics and its current capacity, a set of feasible alternative offers is then generated. From this set, the recommendation strategy selects a limited number of alternatives, which are presented to the customer sequentially. Each offer is evaluated by the customer based on their preference model, and the process ends with the customer accepting one of the offers or rejecting all of the presented alternatives. The outcome of the interaction then serves as feedback for updating the recommendation strategy before processing the next customer.

The individual components of the framework are described in detail in the following subsections.

3.3 Restaurant Model

Restaurant entities represent the supply side of the reservation system. Each restaurant is characterized by a fixed set of operational attributes that remain unchanged throughout a simulation run, while its occupancy calendar is generated separately for every simulated period.

Each restaurant is characterized by five attributes: market segment, opening hours, seating capacity, occupancy calendar, and the set of alternative offers that may be recommended to customers.

Fifteen restaurants are considered in the simulation and divided into three market segments: budget, mid-range, and premium. The segmentation reflects different operating conditions, primarily represented by different baseline occupancy levels. The occupancy calendar of each restaurant is generated independently for every simulated period. The occupancy level depends on the restaurant segment, time of day, and day of the week, while an additional stochastic

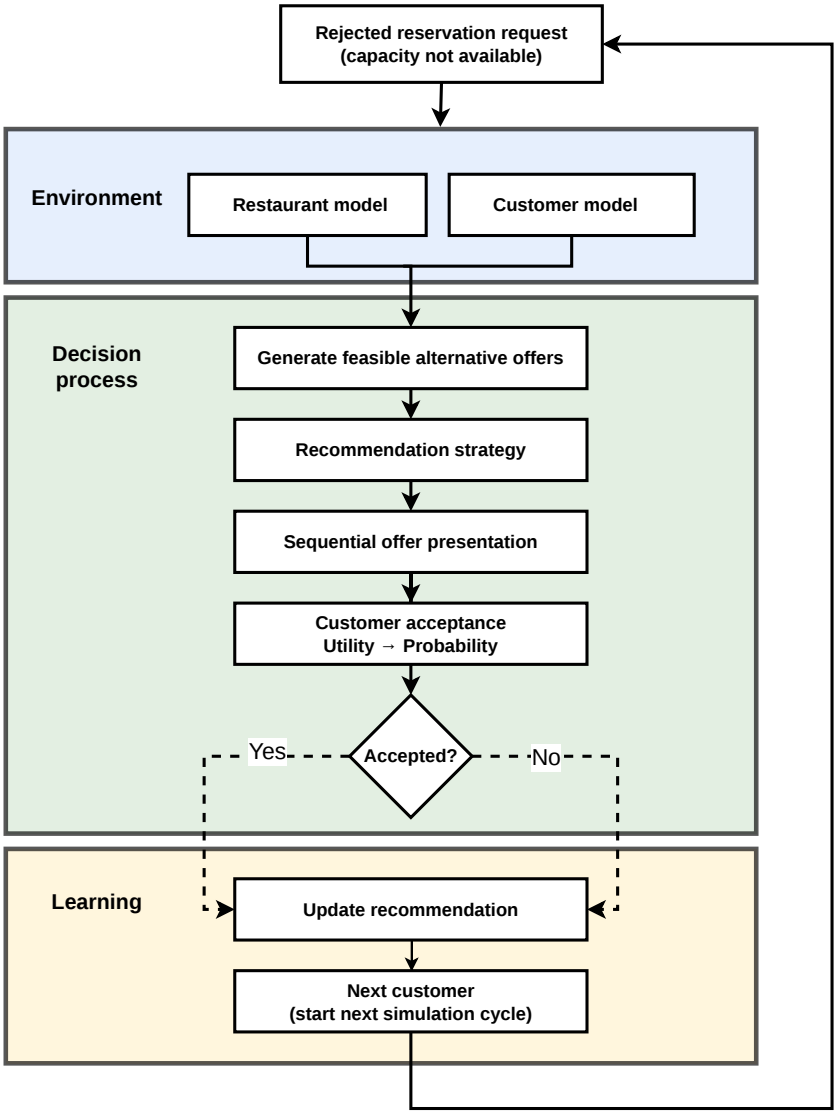


Figure 1: Overview of the proposed simulation framework.

component is introduced to represent natural variability in restaurant demand.

$$O = O_{base} + O_{peak} + O_{weekend} + \varepsilon \tag{1}$$

where

- O_{base} represents the baseline occupancy determined by the restaurant segment,
- O_{peak} denotes the increase in occupancy during evening peak hours,
- $O_{weekend}$ captures the additional demand on weekends,
- $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ represents random variability.

Capacity is represented by the number of available tables in each hourly time slot. An alternative reservation is considered feasible only if sufficient capacity exists for the customer's group size.

3.4 Customer Model

Another component of the simulation framework is synthetic customers representing the demand side of the reservation system. Each customer is characterized by an initial reservation request and a set of individual characteristics that influence their decision-making regarding the acceptance of alternative offers. Customers are generated independently for each simulation run.

The initial reservation request is generated based on assumptions similar to those used for restaurant occupancy. A higher probability of requests is expected in the evening and on weekends. A restaurant is assigned to the customer at random. The model assumes that the initial request has already been rejected; therefore, it does not address the restaurant selection process or capacity verification. The simulation begins only at the stage of offering alternative options.

The reservation request also includes the size of the group being reserved. This affects both the availability of alternative time slots and the customer's willingness to accept a change in the reservation time. The model assumes that groups of five or more people have less time flexibility, which is taken into account through an additional penalty when evaluating alternative offers.

Table 1: Probability distribution of customer group sizes.

Group size	**Probability**
1	15%
2	35%
3	20%
4	12%
5	10%
6	5%
7	3%

Each synthetic customer is further characterized by five individual preferences: discount affinity, time flexibility, restaurant loyalty, perk affinity, and urgency.

Individual characteristics are generated randomly according to a normal distribution and take values in the interval $[0, 1]$. Values close to zero represent a low intensity of a given preference, while values close to one indicate its strong influence on the customer's decision-making. These characteristics are then fed into the model for calculating the utility of individual alternative offers.

3.5 Alternative Offer Model

Alternative offers are the basic decision-making unit of the simulation. Their purpose is to provide customers with an alternative after their original reservation request has been rejected. An offer may combine a change in the reservation time, financial compensation, and a non-financial benefit.

An alternative offer is represented as

$$o = (T, D, P) \in [0, 1]^3, \tag{2}$$

where T denotes the normalized magnitude of the reservation time shift, D denotes the normalized discount offered for the customer's next visit, and P denotes the normalized value of a non-financial benefit. A value of zero indicates that the corresponding component is not included in the offer, while values closer to one represent a larger time shift, a higher discount, or a more valuable non-financial benefit, respectively. The original time shifts and discount percentages are therefore converted to normalized values before being used in the customer acceptance model.

Depending on which components are included, alternative offers can be categorized into several offer types. The categories considered in the simulation are summarized in Table 2.

Table 2: Alternative offer categories.

Offer Category	**Time Shift**	**Discount**	**Perk**
Time Only	Yes	No	No
Discount Only	No	Yes	No
Perk Only	No	No	Yes
Time + Discount	Yes	Yes	No
Time + Perk	Yes	No	Yes
Time + Discount + Perk	Yes	Yes	Yes

3.6 Offer Availability

Before recommendation strategies are applied, the set of alternative offers is filtered to include only those that are feasible under the current operating conditions. The purpose of this filtering process is to remove alternatives that cannot be realistically offered to the customer. Consequently, the recommendation strategy selects offers only from the set of feasible alternatives.

The feasibility of each alternative offer is determined by three consecutive constraints:

1. **Restaurant offer policy.** Only offers supported by the restaurant may be considered. Offers that are not included in the restaurant's predefined offer policy are discarded.
2. **Opening hours.** Alternative reservations requiring customer arrival outside the restaurant's operating hours are removed from the candidate set.
3. **Capacity availability.** An alternative is considered feasible only if sufficient capacity is available in the corresponding time slot for the customer's group size.

The resulting set of feasible offers can be formally expressed as

$$A_f = \{\, o \in A \mid P(o) = 1,\ H(o) = 1,\ C(o) = 1 \,\} \tag{3}$$

where A denotes the catalog of all possible alternative offers, A_f is the set of feasible offers, $P(o)$ indicates that offer o is permitted by the restaurant's offer policy, $H(o)$ denotes compliance with the restaurant's opening hours, and $C(o)$ represents sufficient capacity for the customer's reservation.

The resulting set of feasible alternatives forms the input to the recommendation strategy described in the following section.

3.7 Recommendation Policies

After the set of feasible alternative offers has been determined, the recommendation policy selects at most k offers to be presented to the customer. Since the true probability of customer acceptance is initially unknown, the recommendation problem is formulated as a Multi-Armed Bandit (MAB) problem. Each feasible alternative offer corresponds to one arm, while the observed customer response represents the reward.

The objective of the recommendation policy is to maximize the long-term acceptance rate by balancing the exploration of previously underexplored offers with the exploitation of offers that have demonstrated high acceptance rates. Following each customer interaction, the estimated quality of the selected offers is updated according to the observed reward, allowing the recommendation policy to gradually improve its future decisions.

For all value-based recommendation policies, the estimated value of offer i is computed as

$$Q_i = \frac{S_i}{N_i}, \tag{4}$$

where S_i denotes the cumulative reward obtained by offer i and N_i represents the number of times the offer has been selected.

The simulation compares six recommendation policies.

Random. The Random policy serves as a baseline and selects k feasible offers uniformly at random without using any previous observations.

ϵ-Greedy. The ϵ-Greedy policy balances exploration and exploitation by selecting a random offer with probability ϵ and the offer with the highest estimated value otherwise,

$$a = \begin{cases} \text{random}, & \text{with probability } \epsilon, \\ \arg\max_i Q_i, & \text{otherwise.} \end{cases} \tag{5}$$

Decay ϵ-Greedy. To gradually reduce exploration over time, the Decay ϵ-Greedy policy decreases the exploration probability according to

$$\epsilon_t = \max\left(\epsilon_{\min}, \frac{\epsilon_0}{\sqrt{t}}\right), \tag{6}$$

where t denotes the current interaction step.

UCB1. The UCB1 policy selects offers according to the Upper Confidence Bound criterion,

$$Q_i + c\sqrt{\frac{\ln t}{N_i}}, \tag{7}$$

where c controls the degree of exploration.

Thompson Sampling. Thompson Sampling models the reward probability of each offer using a Beta distribution,

$$\theta_i \sim \text{Beta}(\alpha_i, \beta_i), \tag{8}$$

where the parameters are updated after each customer interaction according to

$$\alpha_i \leftarrow \alpha_i + r, \tag{9}$$

$$\beta_i \leftarrow \beta_i + 1 - r, \tag{10}$$

with binary reward $r \in \{0, 1\}$.

Softmax. The Softmax policy assigns each offer a selection probability proportional to its estimated value,

$$P_i = \frac{\exp(Q_i/\tau)}{\sum_j \exp(Q_j/\tau)}, \tag{11}$$

where τ is the temperature parameter controlling the exploration level.

All recommendation policies operate on the same set of feasible alternative offers and are evaluated under identical simulation settings to enable a fair comparison.

3.8 Customer Acceptance Model

After the recommendation policy selects the alternative offers, the customer decides whether to accept or reject each presented offer. Customer decisions are modeled probabilistically to reflect the inherent uncertainty of human decision-making rather than assuming deterministic behavior.

Each alternative offer is assigned an overall utility value representing its attractiveness to a particular customer. Since the customer has already experienced the rejection of the original reservation request, the model assumes a negative baseline utility. The overall utility is then adjusted according to both the characteristics of the offered alternative and the customer's individual preferences.

The overall utility is computed as

$$U = U_0 + U_{\text{discount}} + U_{\text{perk}} + U_{\text{loyalty}} - U_{\text{delay}} - U_{\text{urgency}} + \varepsilon, \tag{12}$$

where U_0 denotes the baseline utility associated with the rejected reservation request, U_{discount} represents the positive contribution of the offered discount, U_{perk} captures the utility gained from a non-financial benefit, U_{loyalty} reflects the customer's loyalty toward the restaurant, U_{delay} represents the penalty associated with shifting the reservation time, U_{urgency} models the penalty caused by reservation urgency, and ε is a random noise term representing unobserved variability in customer preferences.

The values of the individual utility components are determined by combining the corresponding customer characteristics with the parameters of the offered alternative. The exact parameterization of the utility model is described in the experimental settings.

The resulting utility is transformed into the probability of accepting the offer using the logistic function

$$P(\text{accept}) = \frac{1}{1 + e^{-U}}, \tag{13}$$

which maps utility values onto the interval $[0, 1]$. The final customer decision is obtained by sampling from the corresponding Bernoulli distribution. Consequently, even highly attractive offers may occasionally be rejected, while less attractive offers may still be accepted with a lower probability.

3.9 Sequential Offer Evaluation

The recommendation policy may present up to k alternative offers to a customer. When $k = 1$, only a single offer is evaluated and the customer's decision is based solely on the acceptance probability computed by the customer acceptance model.

For $k > 1$, offers are evaluated sequentially rather than simultaneously. The customer first evaluates the highest-ranked recommendation. If the offer is accepted, the evaluation process terminates and the reservation is considered successfully recovered. Otherwise, the next offer in the recommendation list is presented. This process continues until either one of the offers is accepted or all presented alternatives have been rejected.

Since offers are evaluated sequentially, their ordering directly influences the overall acceptance probability. Consequently, the recommendation policy must determine not only which offers should be presented but also the order in which they are displayed.

After the customer's decision has been completed, the recommendation policy receives feedback that is used to update its internal estimates. If all presented offers are rejected, each evaluated offer receives a reward of zero. If an offer is accepted, the accepted offer receives a reward of one, while all previously rejected offers receive a reward of zero. Offers appearing after the accepted alternative are not evaluated and therefore do not receive any reward update.

3.10 Simulation Cycle

The simulation proceeds as a sequence of customer interactions. During each iteration, one synthetic customer is generated and processed according to the framework described in the previous sections.

For each simulated customer, the following steps are performed:

1. A synthetic customer and a target restaurant are generated.
2. The restaurant occupancy calendar is used to determine the set of feasible alternative offers.
3. The recommendation policy selects up to k offers from the feasible offer set.
4. The selected offers are presented sequentially to the customer.
5. The customer evaluates each offer according to the customer acceptance model.
6. The recommendation policy is updated using the observed customer response.

The above procedure is repeated for all customers within a simulation run. Multiple independent simulation runs are performed for each experimental scenario to account for the stochastic nature of customer generation, restaurant occupancy, and customer decision-making.

4 Experimental Design

This section describes the experimental methodology used to evaluate the proposed simulation framework. First, the baseline simulation settings and model parameters are introduced. Next, the evaluation metrics used to assess recommendation performance are presented. Finally, the experimental scenarios designed to investigate the research questions are described.

4.1 Simulation Settings

Unless stated otherwise, all experiments were conducted using the same baseline simulation configuration. The baseline configuration specifies the default values of all model parameters used throughout the experiments, including the simulation environment, restaurant operating conditions, customer generation process, customer acceptance model, and recommendation policy parameters. Using a common baseline configuration ensures that all recommendation policies are evaluated under identical conditions, enabling a fair comparison of their performance.

Customer preference attributes, including discount affinity, time flexibility, restaurant loyalty, perk affinity, and reservation urgency, are generated from normal distributions with a mean of 0.5 and predefined standard deviations. The customer acceptance model assumes an initial baseline utility of $U_0 = -2.0$, reflecting the negative customer experience caused by the rejection of the original reservation request. To account for the inherent variability of human decision-making, a stochastic noise component with a standard deviation of 0.6 is added to the utility function.

The general simulation settings are summarized in Table 3.

The baseline restaurant environment used throughout the experiments is presented in Table 4. These parameters define the default operating conditions and are modified only in the experimental scenarios specifically designed to evaluate their impact.

The recommendation policies were evaluated using the parameter settings listed in Table 5. Unless explicitly stated otherwise, these values remain unchanged across all experimental scenarios.

Table 3: General simulation settings.

Parameter	**Value**
Customers per simulation run	20 000
Independent simulation runs	100
Random seed	33
Simulation horizon	7 days
Evaluated top-k values	1–7
Baseline utility (U_0)	-2.0
Decision noise (standard deviation)	0.6

Table 4: Baseline restaurant environment parameters.

Parameter	**Value**
Number of restaurants	15
Restaurant segments	Budget, Mid-range, Premium
Opening hours	11:00–23:00
Budget baseline occupancy	45%
Mid-range baseline occupancy	60%
Premium baseline occupancy	75%
Evening occupancy adjustment	+20%
Weekend occupancy adjustment	+15%

Table 5: Recommendation policy parameters.

Policy	**Parameter**	**Value**
ϵ-Greedy	ϵ	0.10
Decay ϵ-Greedy	ϵ_0	0.10
Decay ϵ-Greedy	$\epsilon_{\min}$	0.01
UCB1	c	1.0
Softmax	τ	0.10
Value-based policies	Initial value estimate (Q_0)	0.15

4.2 Evaluation Metrics

The proposed framework is evaluated using three complementary performance metrics. The primary objective is to maximize the proportion of customers who accept at least one recommended alternative. In addition, the experiments analyze the acceptance of different offer categories and compare the performance of adaptive recommendation policies with a random baseline. All reported results correspond to the average over multiple independent simulation runs.

Mean Reward Mean Reward is the primary evaluation metric used throughout the experiments. Each customer interaction produces a binary reward:

$$r_i = \begin{cases} 1, & \text{if the customer accepts at least one alternative offer,} \\ 0, & \text{otherwise.} \end{cases}$$

The Mean Reward is computed as

$$\text{MeanReward} = \frac{1}{N} \sum_{i=1}^{N} r_i,$$

where N denotes the number of simulated customers.

Higher Mean Reward values indicate a higher proportion of successfully recovered reservations and therefore better overall recommendation performance.

Offer Acceptance Rate To analyze the effectiveness of individual offer categories, the Offer Acceptance Rate is computed as

$$\text{AcceptanceRate} = \frac{\text{Number of accepted offers}}{\text{Number of evaluated offers}}.$$

An offer is counted as evaluated only if it is actually presented to the customer. Offers that were selected by the recommendation policy but never displayed because a previous offer was accepted are excluded from the denominator.

This metric is reported separately for each offer category to identify which types of alternatives are most likely to be accepted.

Relative Improvement To quantify the benefit of adaptive recommendation policies over the random baseline, the Relative Improvement is computed as

$$\text{Improvement} = \frac{\text{MeanReward}_{\text{policy}} - \text{MeanReward}_{\text{random}}}{\text{MeanReward}_{\text{random}}} \times 100\%.$$

Positive values indicate an improvement over the random recommendation policy, whereas negative values indicate inferior performance.

4.3 Experimental Scenarios

The proposed framework is evaluated under several experimental scenarios designed to investigate the robustness of the recommendation policies under different operating conditions. All scenarios use the same simulation framework and baseline configuration described in the previous section. Individual scenarios modify only a single model parameter while all remaining parameters remain unchanged.

The evaluated experimental scenarios are summarized in Table 6.

5 Results

5.1 RQ1: Number of Displayed Offers

The first experiment investigates how the maximum number of alternative offers presented to the customer influences the overall recommendation performance. The experiment was conducted using the baseline simulation configuration, while varying the maximum number of sequentially displayed offers from $k = 1$ to $k = 7$. Recommendation performance was evaluated using the Mean Reward metric averaged over all simulation runs.

Figure 2 shows that increasing the number of displayed offers consistently improves the probability that a customer accepts at least one alternative recommendation. Mean Reward increases from 0.1549 for a single displayed offer to 0.5257 when seven alternatives are available.

The largest improvement is observed between $k = 1$ and $k = 4$, where each additional recommendation substantially increases customer acceptance. Beyond this point, the marginal improvement gradually decreases, indicating diminishing returns from presenting additional alternatives. In particular, the difference between six and seven displayed offers is relatively small.

Overall, the results suggest that presenting multiple alternatives considerably improves recommendation effectiveness. However, most of the performance gain is achieved with approximately four to five displayed offers, while additional recommendations provide only limited further benefit.

Table 6: Experimental scenarios.

Experiment	**Scenario**	**Parameter value**
Opening hours	Baseline	11:00–23:00
	Extended	09:00–24:00
Restaurant occupancy (Budget / Mid-range / Premium)	Much Lower	0.15 / 0.30 / 0.45
	Lower	0.30 / 0.45 / 0.60
	Baseline	0.45 / 0.60 / 0.75
	Higher	0.60 / 0.75 / 0.90
	Much Higher	0.75 / 0.90 / 0.98
Decision uncertainty (σ_ε)	Very Low	0.1
	Low	0.3
	Baseline	0.6
	High	1.0
	Very High	1.5
Customer attitude	Very Pessimistic	$U_0 = -3.5$
	Pessimistic	$U_0 = -2.5$
	Baseline	$U_0 = -2.0$
	Optimistic	$U_0 = -1.0$
	Very Optimistic	$U_0 = 0.0$

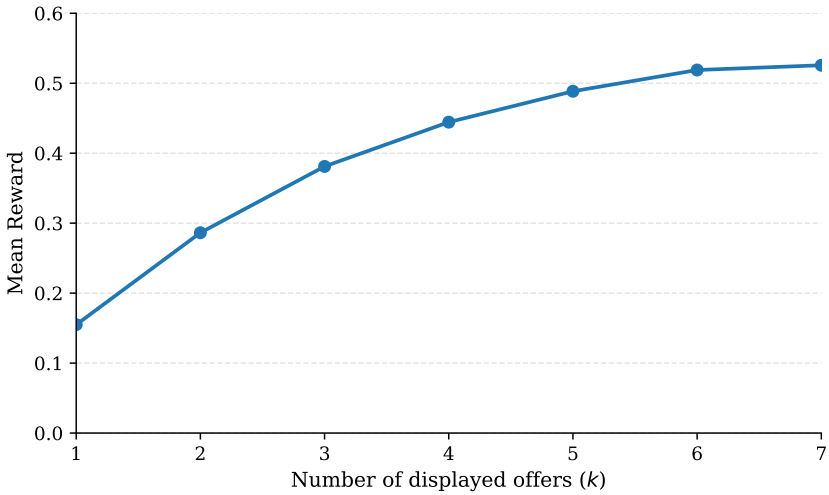


Figure 2: Mean Reward achieved for different values of k, representing the maximum number of alternative offers presented sequentially to the customer.

5.2 RQ2: Alternative Offer Categories

The second experiment investigates how different categories of alternative offers influence customer acceptance. The evaluation was performed using the baseline simulation scenario, while varying the maximum number of displayed offers from $k = 1$ to $k = 7$. For each offer category, the acceptance rate was calculated as the proportion of accepted recommendations among all displayed offers.

Figure 3 compares the acceptance rates achieved by the six offer categories considered in the simulation.

Offers consisting solely of a financial discount consistently achieved the highest acceptance rate across all values of k. The second-best performance was obtained by offers providing only

a non-financial perk, followed by offers combining a reservation time shift, a discount, and a perk. In contrast, offers relying exclusively on a reservation time shift consistently resulted in the lowest customer acceptance.

The relative ordering of offer categories remained stable across all values of k, indicating that customer preferences for different compensation mechanisms are largely independent of the number of displayed alternatives.

A gradual decrease in acceptance rate can be observed for every offer category as the number of displayed offers increases. This behavior is expected because later recommendations are presented only after the customer has rejected all previously displayed offers, making acceptance progressively less likely.

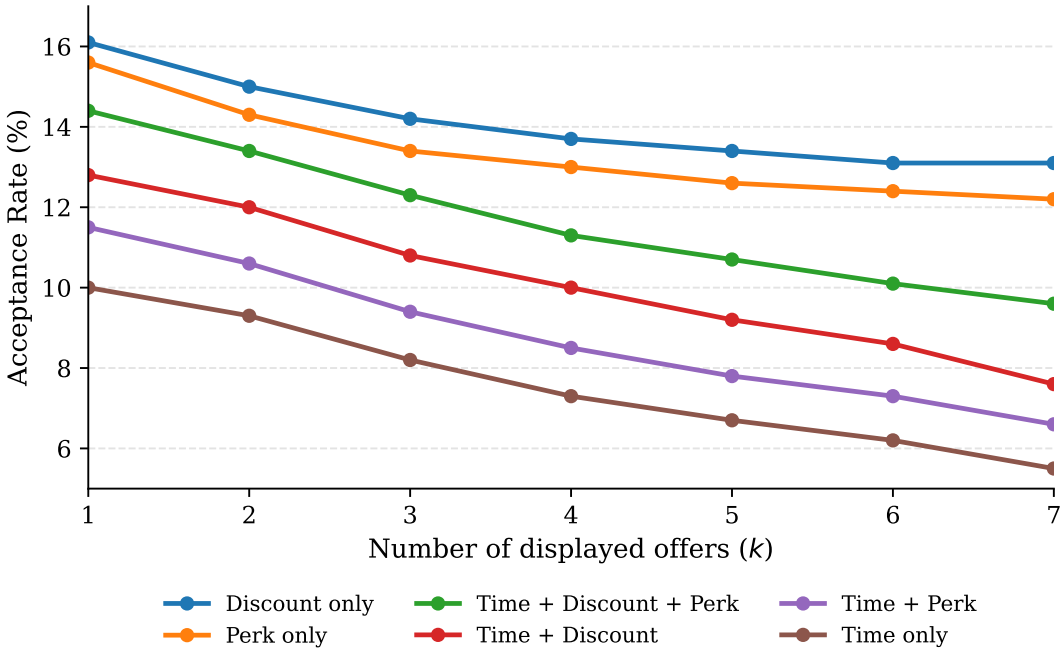


Figure 3: Acceptance rate of individual offer categories for different values of k.

5.3 RQ3: Robustness under Different Operating Conditions

5.3.1 Extended Opening Hours

The first scenario evaluates whether extending restaurant opening hours improves the effectiveness of adaptive recommendation strategies. Compared with the baseline configuration, restaurants remain open for a longer period, increasing the number of feasible reservation time slots that can be offered as alternatives. The experiment evaluates the best-performing recommendation strategy (Decay ϵ-Greedy) and compares its relative improvement over the Random policy.

Figure 4 summarizes the observed relative improvement for different values of k.

The results indicate that extending restaurant opening hours has only a limited influence on recommendation performance. For $k = 1$ and $k = 2$, the improvement over the Random strategy increases by approximately one percentage point. A slightly larger increase is observed for $k = 3$, where the relative improvement rises from 11.76% to 13.57%.

Overall, although additional opening hours increase the number of feasible alternative offers, their impact on the recommendation process remains relatively small under the evaluated simulation settings. This suggests that extending operating hours alone does not substantially improve customer acceptance.

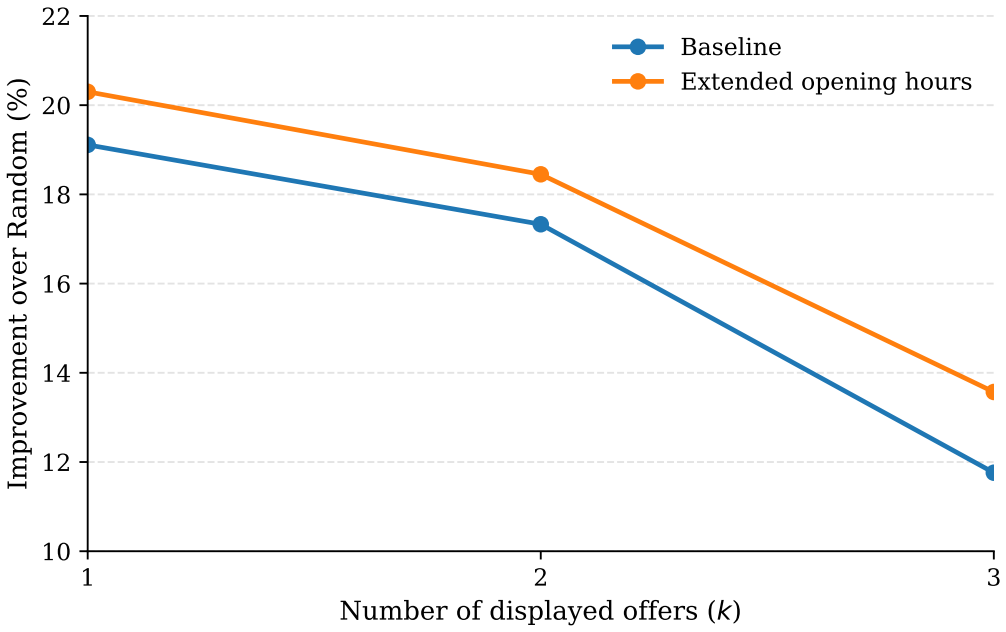


Figure 4: Relative improvement of the Decay ϵ-Greedy strategy over the Random policy under the baseline and extended opening hours scenarios.

5.3.2 Different Occupancy

The second scenario investigates the influence of restaurant occupancy on recommendation performance. Five occupancy scenarios were evaluated, corresponding to the occupancy levels introduced in Section 4.3: *Much Lower*, *Lower*, *Baseline*, *Higher*, and *Much Higher*. The evaluation was performed for $k = 1$ using the best-performing recommendation strategy (Decay ϵ-Greedy).

Figure 5 presents the achieved Mean Reward under the different occupancy scenarios. Only minor differences can be observed across all occupancy levels. Mean Reward gradually increases from 0.1528 in the *Much Lower* scenario to 0.1566 in the *Much Higher* scenario, indicating that restaurant occupancy has only a negligible influence on the absolute performance of the recommendation policy.

In contrast, Figure 6 shows that the *Relative Improvement over Random* changes substantially. The relative advantage of the adaptive recommendation strategy decreases as occupancy increases. While the improvement reaches approximately 22% under lower occupancy conditions, it falls below 9% in the *Much Higher* scenario.

These results indicate that although the absolute recommendation performance remains nearly unchanged, adaptive recommendation strategies provide a considerably larger benefit over random recommendation when restaurant occupancy is lower. As occupancy increases and feasible alternatives become more limited, the advantage of learning-based recommendation gradually diminishes.

5.3.3 Decision Uncertainty

The third scenario evaluates the robustness of the recommendation framework under different levels of stochastic customer behavior. Decision uncertainty is controlled by the standard deviation of the Gaussian noise component added to the utility function. Lower values of the noise parameter result in customer decisions being driven primarily by their individual preferences, whereas higher values introduce a larger random component into the acceptance process. Unless stated otherwise, all other experiments use the baseline value of $\sigma = 0.6$.

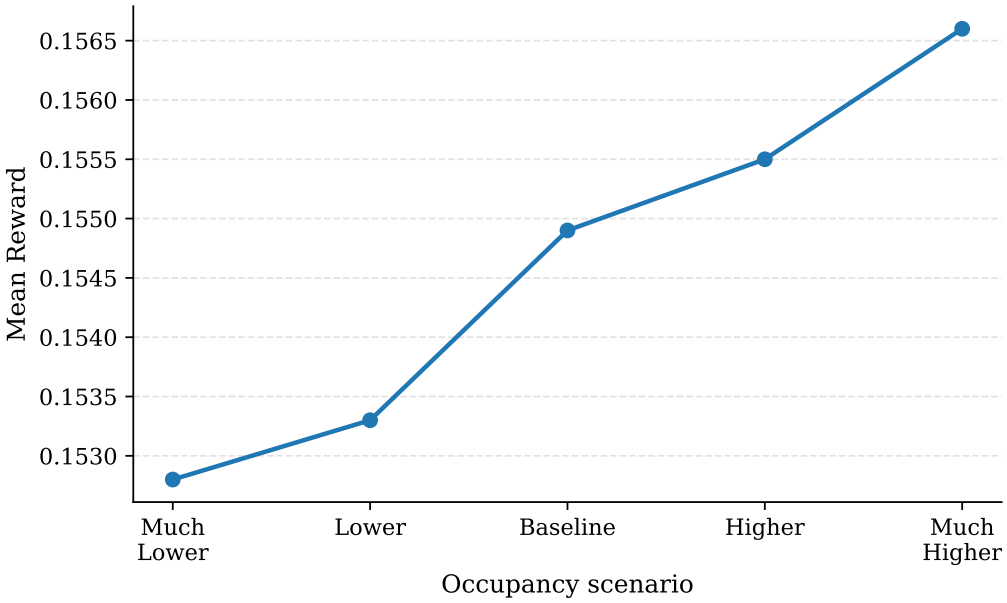


Figure 5: Mean Reward under different occupancy scenarios ($k = 1$).

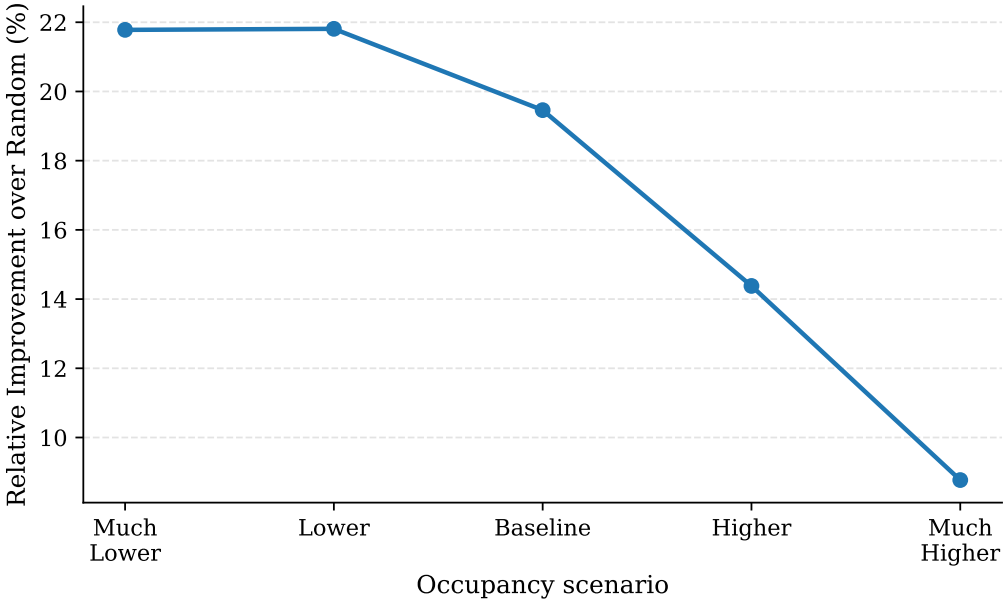


Figure 6: Relative Improvement over Random under different occupancy scenarios ($k = 1$).

Figure 7 presents the Mean Reward achieved by the best-performing recommendation strategy for different levels of decision noise and values of k. As the level of decision uncertainty increases, the Mean Reward consistently improves across all evaluated values of k. The largest absolute increase is observed for $k = 3$, where Mean Reward rises from 0.3500 for $\sigma = 0.1$ to 0.4937 for $\sigma = 1.5$.

The relative performance of the adaptive recommendation strategy compared with the Random baseline is shown in Figure 8. In contrast to the absolute performance, the relative improvement gradually decreases as the decision noise increases. This indicates that although customers become more likely to accept alternative offers under higher levels of randomness, their decisions also become less predictable, reducing the advantage of adaptive learning strategies.

Nevertheless, the adaptive recommendation strategy consistently outperforms the Random

baseline across all tested noise levels, demonstrating the robustness of the proposed approach even under highly stochastic customer behavior.

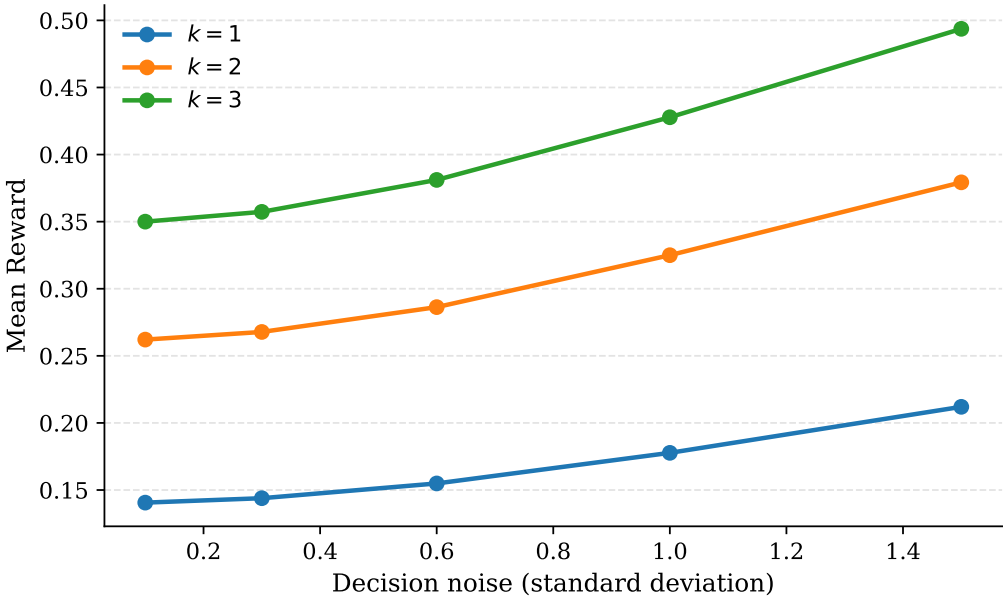


Figure 7: Mean Reward under different levels of decision uncertainty for $k = 1$, $k = 2$, and $k = 3$.

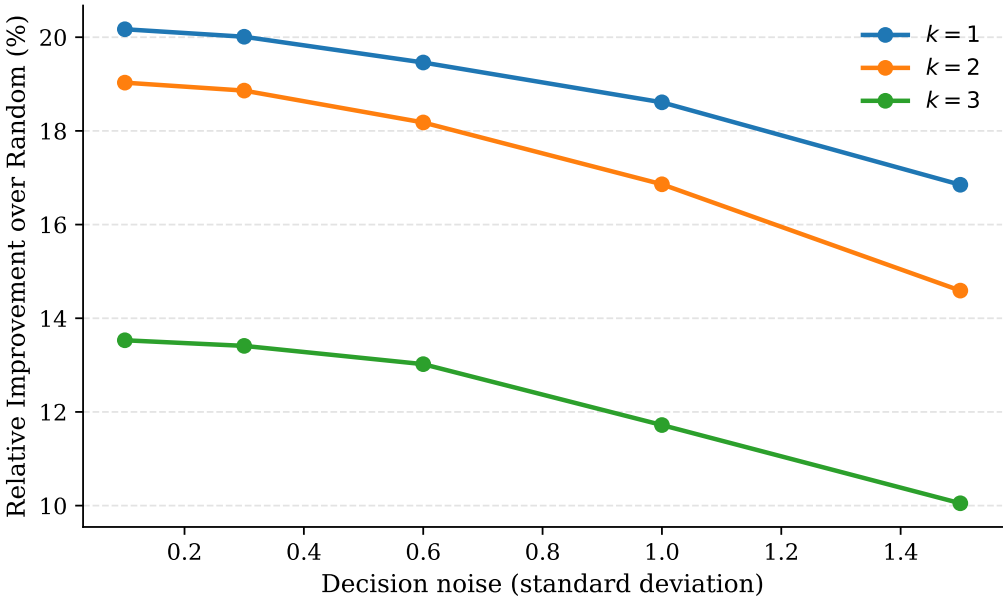


Figure 8: Relative Improvement over Random under different levels of decision uncertainty.

5.3.4 Different Customer Attitudes

The final scenario investigates how the customer's initial attitude toward alternative offers influences recommendation performance. Five customer attitude scenarios were evaluated by modifying the baseline utility parameter U_0, corresponding to the scenarios *Very Pessimistic*, *Pessimistic*, *Baseline*, *Optimistic*, and *Very Optimistic*.

Figure 9 presents the Mean Reward achieved by the best-performing recommendation strategy for different customer attitude scenarios and values of k. As expected, recommendation performance increases substantially as customers become more willing to accept alternative offers. For $k = 1$, Mean Reward increases from 0.0398 in the *Very Pessimistic* scenario to 0.5444 in the *Very Optimistic* scenario. Similar trends are observed for $k = 2$ and $k = 3$, where the achieved Mean Reward increases from 0.0793 to 0.7914 and from 0.1119 to 0.8903, respectively. For $k = 3$, the Mean Reward in the most optimistic scenario is nearly eight times higher than in the most pessimistic scenario.

Figure 10 shows the Relative Improvement over Random for the same scenarios. Unlike the absolute performance, the relative advantage of adaptive recommendation is highest around the *Baseline* and *Pessimistic* scenarios. As customers become increasingly optimistic, the advantage over random recommendation gradually decreases, particularly for larger values of k. The lowest relative improvement is observed in the *Very Pessimistic* scenario, while the most pronounced decline with increasing k occurs under the *Very Optimistic* setting.

These results indicate that customer attitude has a substantial influence on the absolute effectiveness of the recommendation system. However, the relative benefit of adaptive recommendation strategies is greatest for customers whose willingness to accept alternative offers is neither extremely low nor extremely high.

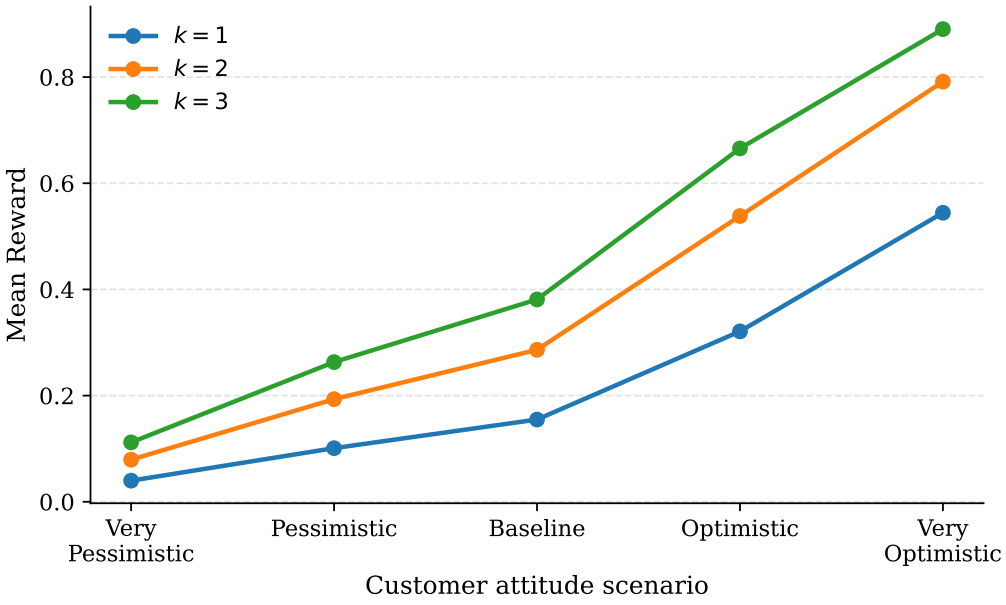


Figure 9: Mean Reward under different customer attitude scenarios for $k = 1$, $k = 2$, and $k = 3$.

5.4 RQ4: Comparison of Recommendation Strategies

The final experiment compares the performance of the evaluated recommendation strategies under the baseline simulation configuration. The Random policy serves as the baseline, while the remaining strategies correspond to the adaptive Multi-Armed Bandit algorithms introduced in Section 2.4. The comparison considers both the relative improvement achieved over the Random policy and the differences among the individual adaptive strategies for different values of k.

Figure 11 summarizes the relative improvement achieved by the adaptive recommendation strategies. All evaluated learning-based approaches outperform random recommendation for small values of k, confirming the benefit of adaptive offer selection. This advantage is most pronounced for $k = 1$, when the recommendation policy must select a single alternative from

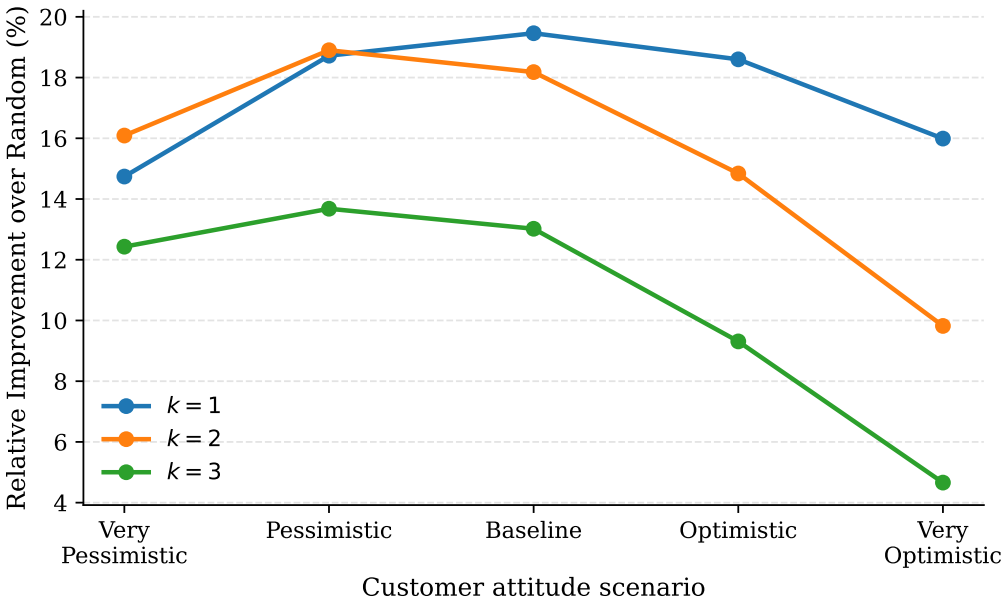


Figure 10: Relative Improvement over Random under different customer attitude scenarios.

the feasible candidate set.

A notable result is that the differences among the evaluated adaptive strategies are comparatively small. Decay ϵ-Greedy achieves the highest numerical improvement over the Random baseline, reaching approximately 19.5% for $k = 1$. However, its performance is closely followed by ϵ-Greedy and Thompson Sampling across the evaluated values of k. UCB1 provides a somewhat smaller but stable improvement, while Softmax achieves the lowest relative improvement among the adaptive strategies. Overall, the results do not reveal a substantial performance advantage of any single Multi-Armed Bandit algorithm over the other adaptive approaches.

As the number of displayed offers increases, both the advantage over random selection and the differences among the adaptive strategies gradually diminish. For $k = 7$, all strategies achieve nearly identical performance, with relative improvements close to zero. This behavior is expected because presenting a larger number of alternatives reduces the influence of the recommendation policy, as customers eventually receive nearly all feasible offers regardless of their ordering.

Taken together, these findings suggest that the main benefit comes from using an adaptive recommendation strategy rather than random offer selection, rather than from choosing a particular standard Multi-Armed Bandit algorithm. Although Decay ϵ-Greedy achieves the best numerical performance under the baseline configuration, its advantage over the other adaptive strategies remains limited.

When considered together with the results for RQ1 and RQ2, the comparison further suggests that recommendation performance is influenced more strongly by the number, type, and composition of the available alternatives than by the choice among the evaluated adaptive algorithms. Within the proposed simulation framework, improvements in candidate offer generation and personalization may therefore provide greater benefits than replacing one standard Multi-Armed Bandit strategy with another.

6 Discussion

6.1 Main Findings

This study investigated the use of adaptive recommendation strategies for generating alternative restaurant reservation offers through a simulation framework. The experimental results provide

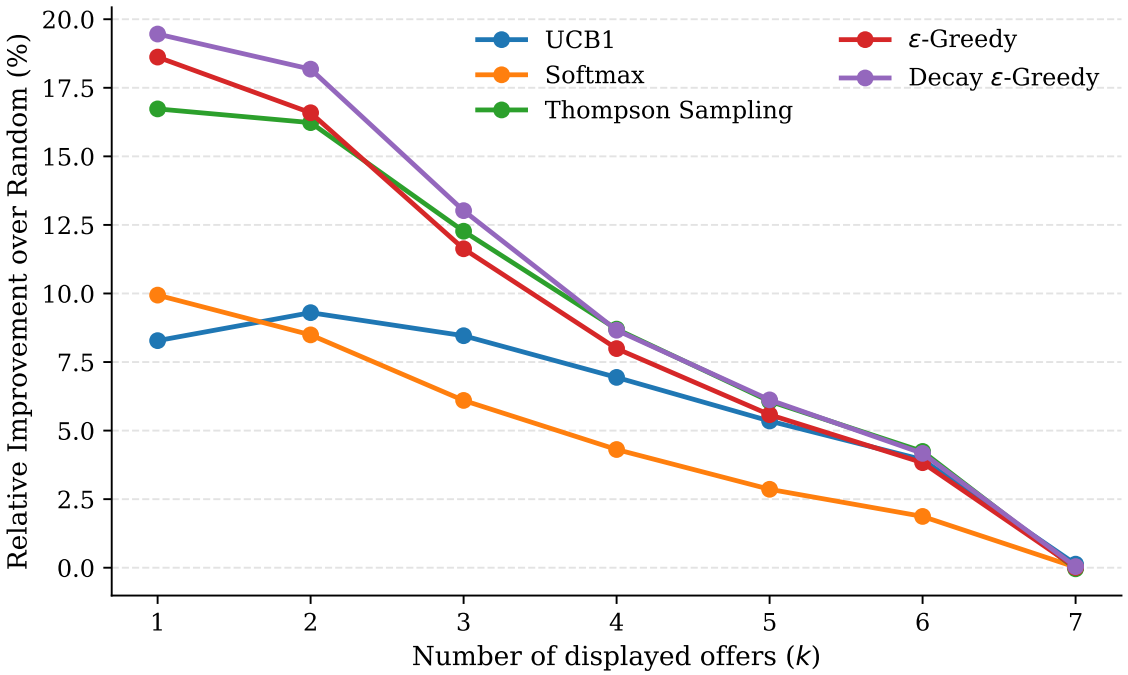


Figure 11: Relative improvement over the Random recommendation policy achieved by the evaluated adaptive strategies for different values of k.

several insights into the factors influencing recommendation performance and the effectiveness of adaptive recommendation policies.

First, the number of displayed alternative offers was identified as the most influential factor affecting customer acceptance. Increasing the number of presented alternatives consistently improved the probability of recovering an unsuccessful reservation. However, this relationship was not linear. The largest improvement was observed when increasing the number of displayed offers from one to four, whereas additional alternatives provided only marginal gains. From the perspective of restaurant reservation systems, these findings suggest that presenting a carefully selected subset of alternatives may be more beneficial than displaying the largest possible number of available offers.

The observed saturation is partly explained by the design of the proposed simulation framework. As the number of displayed alternatives increases, the probability of finding additional feasible offers gradually decreases due to restaurant capacity constraints. Furthermore, offers are evaluated sequentially and the evaluation process terminates immediately after the first accepted alternative, naturally reducing the contribution of lower-ranked recommendations.

The experiments further demonstrated that the type of alternative offer substantially influences customer acceptance. Offers containing financial compensation consistently achieved the highest acceptance rates, whereas alternatives relying solely on reservation time shifts produced the lowest performance. One possible explanation is that changing the reservation time may simply not be feasible for many customers. In contrast, financial compensation represents a more universally applicable incentive because it does not require customers to modify their original schedule or coordinate changes with other members of their reservation group. This observation is consistent with the principles of restaurant revenue management, where pricing incentives are commonly employed to influence customer demand and encourage acceptance of alternative reservation options.

The experimental scenarios further demonstrated that the effectiveness of adaptive recommendation is highly dependent on the characteristics of the decision environment. While operational changes such as extended restaurant opening hours produced only limited effects on

overall system performance, restaurant occupancy, customer decision noise, and customer attitude substantially affected the relative advantage of adaptive recommendation over random offer selection. Across all evaluated scenarios, adaptive recommendation achieved the greatest benefit when sufficient variability existed between available alternatives. Conversely, environments characterized by limited feasible offers or highly stochastic customer decisions naturally reduced the opportunity for learning algorithms to distinguish between better and worse recommendations. This observation is consistent with previous studies on Multi-Armed Bandit algorithms, which emphasize that algorithm performance depends not only on the learning strategy itself but also on the properties of the underlying environment.

Finally, the comparison of recommendation strategies confirmed that adaptive learning consistently outperformed random recommendation. Although all evaluated Multi-Armed Bandit algorithms achieved higher acceptance rates than the Random baseline, the differences between individual adaptive strategies remained relatively small throughout most experimental scenarios. This suggests that, within the proposed simulation framework, the quality and diversity of generated alternative offers have a greater influence on overall system performance than the selection of a particular learning algorithm.

Overall, the proposed simulation framework should primarily be viewed as a tool for analyzing the interaction between alternative reservation offers, restaurant operating conditions, and customer behavior. The results indicate that future improvements in recommendation performance are more likely to be achieved through the design and personalization of alternative offers than through replacing one adaptive recommendation strategy with another.

6.2 Managerial Implications

The proposed simulation framework provides restaurant operators and reservation platform developers with a tool for evaluating alternative reservation recommendation strategies before their deployment in real-world systems. By simulating different operating conditions and customer populations, the framework enables systematic comparison of alternative recommendation policies without requiring costly field experiments.

The experimental results indicate that adaptive recommendation strategies consistently outperform random offer selection. However, the observed differences between individual Multi-Armed Bandit algorithms are relatively small. From a practical perspective, this suggests that greater improvements in reservation recovery may be achieved through the design and personalization of alternative offers than through selecting a particular adaptive recommendation algorithm. Consequently, practitioners may benefit more from investing in the quality of generated offers than from optimizing the learning strategy itself.

The study also highlights the importance of selecting appropriate compensation mechanisms when the originally requested reservation cannot be accommodated. Within the proposed simulation environment, offers containing financial compensation consistently achieved the highest customer acceptance rates. This finding suggests that financial incentives may represent an effective mechanism for encouraging customers to accept alternative reservation options while reducing the loss of potential reservations.

6.3 Limitations

The proposed simulation framework represents a simplified abstraction of the alternative reservation recommendation process in restaurant reservation systems. Although the model parameters were derived from previous literature and selected to approximate realistic operating conditions, the obtained results should not be interpreted as direct predictions of real customer behavior. Rather, the findings are valid within the assumptions and structure of the proposed simulation environment. The proposed framework has been evaluated exclusively using synthetically generated restaurants, customers, and reservation requests. Consequently, although the simu-

lation enables controlled experimentation, further validation using real-world reservation data is necessary before drawing conclusions about deployment in operational restaurant reservation systems.

Another important limitation concerns the customer decision model. Customer behavior is represented using a utility-based model with predefined preference attributes and decision parameters. In practice, customer decisions may also be influenced by many additional factors, including situational context, social influences, previous experiences with the restaurant, and subjective perceptions of different compensation types. Furthermore, the current framework assumes that customer preferences remain constant throughout the simulation and do not evolve as a consequence of previous interactions.

The evaluation criterion also represents a limitation of the present study. The proposed recommendation strategies are optimized solely with respect to the customer acceptance rate, without considering the economic consequences of different compensation policies for restaurant operators. From a business perspective, maximizing acceptance does not necessarily correspond to maximizing restaurant revenue or profitability. Future simulation models should therefore consider multi-objective optimization, incorporating both customer acceptance and the expected financial impact of alternative reservation strategies.

Despite these limitations, the proposed framework provides a flexible environment for systematically evaluating alternative reservation recommendation strategies under a wide range of operational conditions. Consequently, the presented results should primarily be regarded as a foundation for further model refinement and subsequent validation using real-world reservation data.

6.4 Future Work

The proposed simulation framework provides several opportunities for future research. One promising direction is the integration of contextual multi-armed bandits, which would incorporate additional contextual information into the recommendation process. While the current model learns a single recommendation policy from historical customer interactions, contextual bandits estimate the probability of acceptance conditioned on both the offered alternative and the surrounding context,

$$P(\text{accept} \mid \text{offer}, \text{context}).$$

Rather than relying on unobservable customer characteristics, the contextual information could include variables such as restaurant occupancy, requested reservation time, day of the week, time of day, and customer information available through the reservation or registration process. An important research question is whether incorporating contextual information can improve alternative reservation recommendations compared with non-contextual multi-armed bandit approaches.

Another limitation of the current framework is the assumption that customer preferences remain constant throughout the simulation. In practice, customer behavior is likely to change over time due to seasonality, holidays, changing restaurant conditions, economic factors such as inflation, or previous interactions with the reservation system. Future work could therefore investigate non-stationary recommendation environments in which customer preferences evolve over time and recommendation policies continuously adapt to these changes.

Future research should also consider more realistic reward formulations. In the current framework, rewards are binary and depend solely on whether the customer accepts or rejects the proposed alternative. However, accepting an alternative reservation does not necessarily maximize restaurant performance. A more realistic objective would evaluate recommendations using realized customer value, for example by considering whether the customer ultimately visits the restaurant and defining the reward as the resulting revenue after accounting for discounts or complimentary benefits. Such an extension would allow the recommendation process to optimize

business outcomes rather than acceptance rates alone and could be evaluated using data collected from real restaurant reservation systems.

Another promising direction is the integration of active learning techniques into the recommendation process. Rather than relying exclusively on customer acceptance or rejection, the system could actively acquire additional information about customer preferences whenever the expected informational value outweighs the interaction cost. Such an approach could accelerate personalization while reducing the amount of interaction data required to learn effective recommendation policies.

Finally, although the results indicate that displaying a larger number of alternatives generally increases acceptance rates, the current simulation does not consider potential behavioral effects associated with presenting many options. Future work could therefore incorporate mechanisms representing the phenomenon of choice overload, which may influence customer decision-making when a larger number of reservation alternatives is displayed.

7 Conclusion

This study addressed the challenge of retaining customers whose original restaurant reservation requests could not be accommodated because of capacity constraints. The primary objective was to develop a simulation framework for evaluating alternative reservation offers selected using adaptive recommendation strategies. The proposed framework was evaluated across several simulation scenarios representing different restaurant operating conditions and patterns of customer decision-making.

Overall, adaptive recommendation strategies outperformed random selection, particularly when only a small number of alternatives was displayed. However, the differences among the evaluated Multi-Armed Bandit algorithms remained comparatively small across most simulation scenarios. Although Decay ϵ-Greedy achieved the highest numerical performance under the baseline configuration, including an approximately 19.5% relative improvement over random selection for $k = 1$, its advantage over the other adaptive strategies was limited. The primary benefit therefore appears to come from using adaptive rather than random offer selection, rather than from identifying a single universally superior Multi-Armed Bandit algorithm.

The design and composition of the alternative offer set had a greater influence on reservation recovery. Increasing the number of displayed alternatives from one to seven raised the recovery rate from 15.49% to 52.57%, with most of the improvement occurring within the first four to five offers. The type of alternative also played an important role, with offers involving financial incentives achieving the highest acceptance rates. Restaurant occupancy, customer decision uncertainty, and customers' initial attitudes further influenced the relative benefit of adaptive recommendation.

These findings suggest that restaurant reservation systems should prioritize the generation of feasible, relevant, and sufficiently diverse alternatives and subsequently apply an adaptive strategy to select and order them. Once an adaptive approach is used, further improvements in reservation recovery may be more likely to result from improving and personalizing the available alternatives than from replacing one standard Multi-Armed Bandit algorithm with another.

More broadly, the proposed framework enables restaurant operators and reservation platforms to evaluate not only which recommendation strategy performs best, but also whether the differences among algorithms are practically meaningful relative to the effects of offer design, operating conditions, and customer behavior. The framework can therefore support the evaluation of recommendation strategies and alternative offer designs before their potential deployment in operational restaurant reservation systems. However, because the findings were obtained within a simulated environment, validation using real-world reservation data is required before they can be generalized to real customer behavior.

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Generative AI disclosure

During the preparation of this manuscript, the authors used OpenAI ChatGPT (GPT-5, accessed March-August 2026) for English language editing, improving textual clarity and consistency, assisting with the preparation of code documentation, and code refactoring. The tool was not used to generate research data, conduct the analysis, or determine the results or conclusions of the study. All AI-assisted output was critically reviewed, verified, and revised by the authors, who take full responsibility for the content of the manuscript and the accuracy of its references.

Data availability statement

The source code, simulation configuration, and materials required to reproduce the experiments presented in this study are publicly available at `https://github.com/Alzpeta/MAB`. All data analyzed in this study were generated synthetically using the simulation framework.

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